

New open source FPGA development board released on GitHub

The new FPGA development board, ULX3S, is available for download on GitHub. The board was designed and manufactured in Croatia as a result of the cooperation between Radiona, the faculty of the electrical engineering and computing (FER) department of the University of Zagreb, and the Končar-INEM Company.

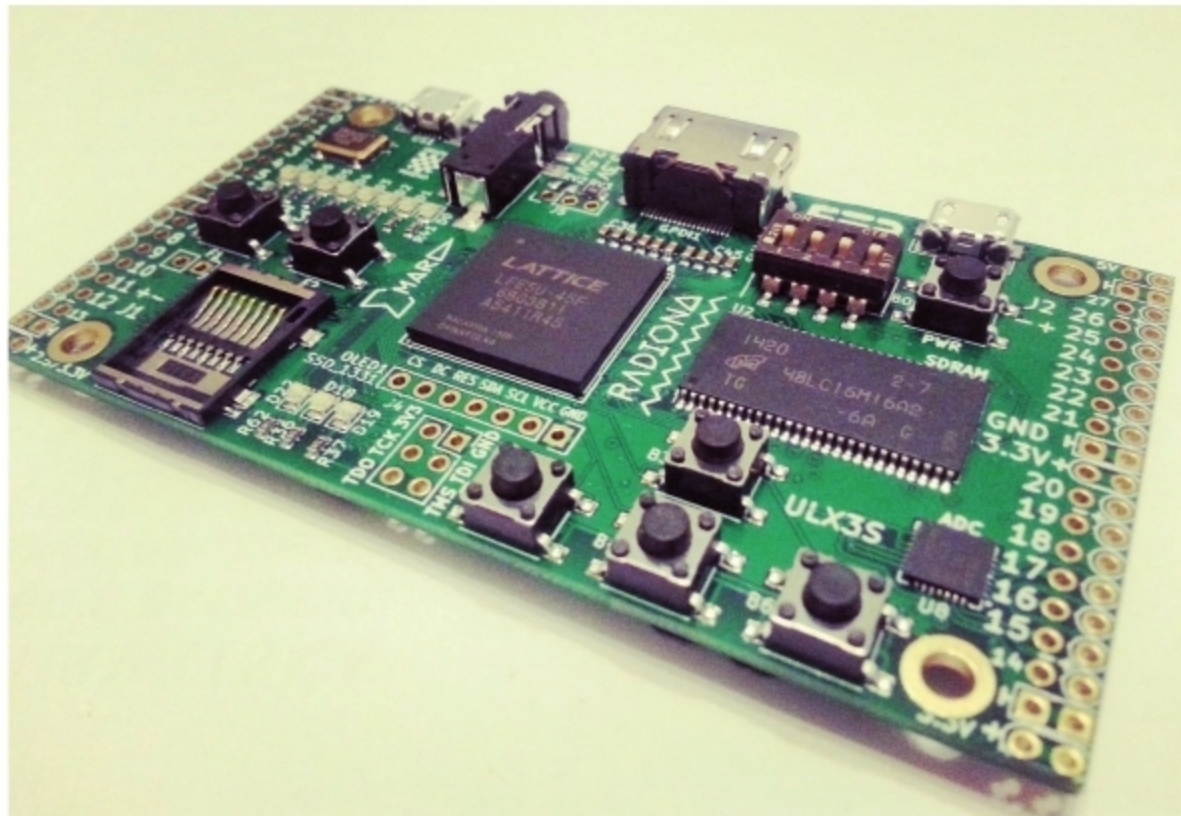
The ULX3S, university digital logic learning Xtensible board, release 3 (with SDRAM), is the successor of ULX2S. It is designed for the development and testing of digital circuits and systems that can be synthesised in integrated field programmable gate array (FPGA) modules.

It was primarily developed as a teaching tool to master the working, analysis and design of digital circuits, but because of its small dimensions and the possibility of connections with external circuits, it can also serve as a broad-spectrum module for installation in complex devices.

“The ULX3S board was designed by Radiona member Davor Jadrijević in response to the needs for a compact, robust and affordable FPGA board equipped with a balanced spectrum of embedded additional components and expansions, while fully covering the requirements of the Digital Logic Course at FER,” a blog post on radiona.org says.

The board is completely open source and its source code is now available on GitHub. The KiCAD 5 design files for the PCB have been released under an MIT-style licence.

The description on GitHub reads as follows: “This is a small (94mm×51mm) standalone FPGA board for education, research and general purposes, with a full-featured selection of chips which are all wanted on a low-cost PCB, but haven’t been found on the market. It is open source and open hardware because if it is good for our development of f32c SOC, it should also be good for other projects.”

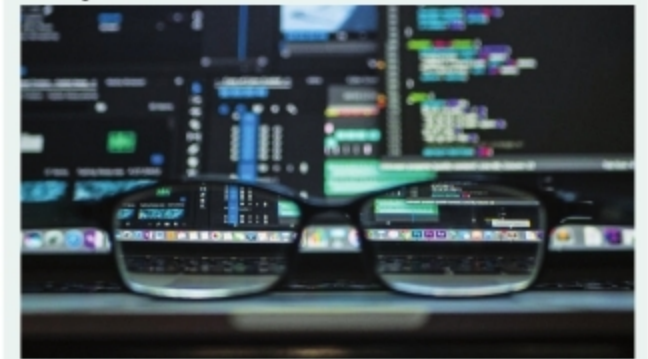


The designers claim ULX3S has the rich range of chips and connectivity required for many other open source projects—SDRAM, USB, PS/2, SDCARD, HDMI, FLASH, RTC, I2C, OLED, AUDIO, ADC, onboard FM/ASK antenna 88-433.92MHz and 56 GPIO pins, which are all routed as differential pairs and have a pin-out that is PMOD (peripheral module) compatible.

EU to fund bug bounties for open source software

The European Union (EU) is looking for bug hunters to discover security flaws in some of the well-known free and open source software.

The EU will fund a total of 15 ‘bug bounties’, of which 14 were launched in January and the remaining one in March this year.



The list of open source projects includes 7-zip, Apache Tomcat, Drupal, Filezilla, VLC, KeePass, Notepad++ and other popular tools on which EU institutions rely. The bug hunters will be offered rewards ranging from US\$ 28,600 to US\$ 103,000, from a total pool amounting to US\$ 973,000.

The amount of the bounty will depend on the severity of the issue uncovered and the relative importance of the software.

The funding will be provided through the EU’s Free and Open Source Software Audit (FOSSA) project, which aims to ensure the integrity and reliability of the Internet and other infrastructure.

The FOSSA project was started in 2014 after security vulnerabilities were discovered in the OpenSSL encryption library. Since then the project has been gathering data, sponsoring hackathons, and deciding on which bug bounties to offer.

The first phase of the project focused on auditing the security of the essential Apache and KeePass software; then the project was renewed and extended to cover other open source software as well.

The bug bounty programme was announced recently by Julia Reda, member of the European Pirate Party and co-founder of the FOSSA project, in a blog post on her website.